

LINEAR ALGEBRA: ALTERNATE COORDINATES

ROWSPACE

The span of all rows of A , where each row is a vector of $\mathbb{R}^n$.

*Subspace iff

$$\vec{a}, \vec{b} \in V^\perp$$

$$\vec{a} + \vec{b} \in V^\perp$$

$$c\vec{a} \in V^\perp$$

NULL SPACE

All vectors $\vec{x}$ where $A\vec{x} = 0$

Nullspace: Information "lost" in transform

eg: A is $\mathbb{R}^2$. Mapping to $\mathbb{R}^3$, null space is everything not on the plane.

What the matrix ignores.

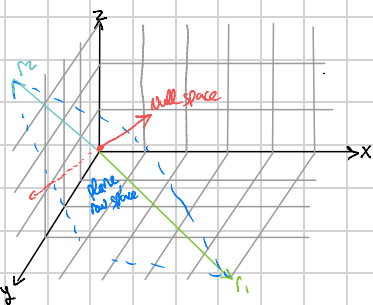
EXAMPLE:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{matrix} \leftarrow r_1 \\ \leftarrow r_2 \end{matrix} \left. \begin{matrix} \text{plane in } \mathbb{R}^3 \\ \text{rowspace} \end{matrix} \right\}$$

$\vec{x}$: nullspace:

$$\begin{aligned} a + b &= 0, \quad b = -1, \quad a = 1, \quad c = 1: \quad \vec{x} = (1, -1, 1) \\ b + c &= 0 \end{aligned}$$

(1 to both r_1, r_2)

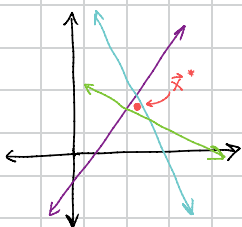


LEAST SQUARES EST

In a system w. no soln. Find the closest possible approximation:

$$A^T A \vec{x}^* = A^T \vec{b}$$

our approx.



ORTHOGONAL COMPLEMENT

Given subspace V in $\mathbb{R}^n$,

$$V^\perp = \{ \vec{x} \in \mathbb{R}^n : \vec{x} \cdot \vec{v} = 0 \text{ for all } \vec{v} \in V \}$$

a subspace of dim k : $V^\perp$ has dim $n-k$.

(V is plane: $V^\perp$ is line)
(V is line: $V^\perp$ is point)

Orthogonal Complements can split a system into 2 parts, so we can:

→ project onto subspaces

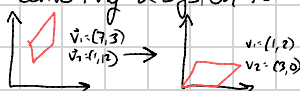
→ decompose vectors

→ build orthonormal bases

→ solve least-squares

ALTERNATE BASES

By converting a system to a new basis, simplify:



Do more math up front to save computation later

$$\text{Find } D \text{ where } T(\vec{x})_B = D \vec{x}_B$$

Orthonormal Bases: Great coordinate systems! Easy to project

→ $\|\vec{v}_i\| = 1$ for all entries ($\vec{v}_i \cdot \vec{v}_i = 1$)

→ $\vec{v}_i \cdot \vec{v}_j = 0$ for $i \neq j$ (all $\vec{v}$ orthogonal to one another)

Gram-Schmidt Process: Constructing & normalizing vectors to make orthonormal bases

SUBSPACE LIN. PROJECTION

A projection onto a subspace splits $\vec{x}$ into $\vec{x}_\parallel + \vec{x}_\perp$

$$\text{proj}_V(\vec{x}) = \vec{x}_\parallel$$



EIGENVALUES, VECTORS

$T(\vec{v}) = \lambda \vec{v}$: $\vec{v}$ is eigenvector (stretch) and λ is eigenvalue (scale)

λ is an eigenvalue if $\det(\lambda I - A) = 0$

Ex) Finding $\vec{v}$ for λ : solve $\lambda I - A = \vec{0}$:

$$\lambda I - A = \vec{0} \rightarrow \begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}: 1\vec{v}_1 - \frac{1}{2}\vec{v}_2 + \frac{1}{2}\vec{v}_3 = 0 \rightarrow \vec{v}_1 = \frac{1}{2}\vec{v}_2 - \frac{1}{2}\vec{v}_3$$

Example

$$\hookrightarrow E_{\lambda=0} = \left\{ \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} : a \begin{bmatrix} v_1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} v_2 \\ 0 \\ 1 \end{bmatrix} \right\} = \text{span} \left(\begin{bmatrix} a \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} b \\ 0 \\ 1 \end{bmatrix} \right)$$